Shameer Sheikh

Tampa, FL · Seeking Full-Time Software Engineering Roles Starting Jan. 2027 · Open To Relocation

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Education

University of South Florida — *Tampa, Florida*

Bachelor of Science, Computer Science | Minor in Psychology — *Aug. 2022 – Expected Dec. 2026*

- **GPA: 3.6/4.0 | USF Green & Gold Presidential Scholar**
- **Relevant Coursework**: Data Structures and Algorithms, Generative AI, Natural Language Processing, Computer Vision, Database Design, Computer Networks, Operating Systems, Software Engineering, Computer Architecture

Technical Skills

Programming Languages, Java, Python, C, JavaScript, HTML, CSS, SQL
Frameworks & Libraries, Spring Boot, React, Next.js, FastAPI, PyTorch, NumPy, OpenCV, MediaPipe
Tools & Databases, PostgreSQL, MongoDB, Redis, Git, Docker, REST APIs, OAuth 2.0, Jupyter, Jira

Work Experience

Software Engineer Intern — *Tampa, Florida*

Ummah Professionals | JavaScript, Next.js, Supabase — *Jun. 2026 – Aug. 2026*

- Built a three-stage **recommendation system** for student-advisor matching: **candidate generation** via local pre-scoring (60+ advisors → top 15), **LLM**-based career-fit classification on the shortlist, and a weighted 5-factor ranking (career fit, industry, experience, gender preference, capacity) producing top-5 recommendations.
- Cut recommendation generation time from ∼**60s to 3–5s** and reduced **LLM** API calls by **75%** by restricting parallel **Gemini** (via OpenRouter) calls to a pre-scored shortlist, with career-fit scores **cached** to stay under API rate limits.

Software Engineer Intern — *Tampa, Florida*

SamSearch | Python, React.js, FastAPI, PostgreSQL, OAuth 2.0, Redis/RQ — *Mar. 2026 – May. 2026*

- Shipped **HubSpot, Salesforce, and Asana integrations** into an **AI-powered government contract intelligence platform** used by **100+ contractors**, implementing **OAuth 2.0 authorization-code flows** across three providers with org-level multi-tenant token storage.
- Introduced **Redis + RQ** to the backend to route all third-party API calls through three dedicated worker processes, enqueuing jobs from **CRUD** events gated on per-org JSONB config toggles to prevent rate-limit cascades and keep external calls off the request path.

Research Assistant — *Tampa, Florida*

Secure and Intelligent Systems Lab, University of South Florida | Python, OpenCV, NumPy — *Jan. 2026 – Mar. 2026*

- Built **preprocessing pipelines** turning **1TB+** of maritime vessel footage (**1,000+ hrs**) into structured face-crop datasets using **YuNet face detection**, supporting crew-identification and work-hour compliance research under Dr. Yasin Yilmaz.

Research Assistant — *Tampa, Florida*

Center for Urban Transportation Research, University of South Florida | Python — *Apr. 2026 – May. 2026*

- Built **ground-truth datasets** across **80** field sites and **1,000+** micro-mobility devices (helmet use, direction, demographics) via synchronized video and digital logging, used to **train and evaluate computer-vision models** for transportation-safety research.

Engineering Projects

LLM Robustness Research — *Tampa, Florida*

NLP Researcher | Groq API, Llama-3.3-70B, Llama-3.1-8B, Qwen3-32B, McNemar's Test, Chain-of-Thought, Self-Consistency — *Mar. 2026 - May. 2026*

- Evaluated whether **Chain-of-Thought** and **Self-Consistency** prompting recover LLM accuracy degraded by irrelevant context on the **GSM-Symbolic** benchmark, running **4,200+** inference calls across **Llama-3.3-70B, Llama-3.1-8B, and Qwen3-32B**.
- Defined a recovery-rate metric showing Self-Consistency **fully closes the accuracy gap** on **Llama-3.3-70B**; annotation of **30 CoT traces** showed models never explicitly misuse the irrelevant sentences despite the accuracy drops.

Generative AI Movie Recommendation System — *Tampa, Florida*

ML Engineer | Python, Groq/Llama-3.1-8B, FAISS, Sentence-Transformers, Pandas, NumPy — *May. 2026*

- Built a **GenAI recommendation system** that uses an **LLM** to **verbalize** each user's watch history into a **natural-language taste profile**, then **retrieves candidates** with **FAISS** and **reranks** them with the **LLM**.
- Evaluated on **MovieLens-1M** (all **6,034** users, leave-last-out), nearly doubling catalog coverage (0.23 → 0.44) over an **embedding-similarity** baseline while surfacing recommendations across the long tail of the **3,883**-movie catalog.

American Sign Language (ASL) Recognition System — *Tampa, Florida*

ML Engineer, Team of 4 | Python, CNN, MediaPipe, OpenCV, Computer Vision, Machine Learning — *Nov. 2025 - Dec. 2025*

- **Developed** a **real-time ASL recognition system** for **36** gestures (A–Z, 0–9), implementing a **custom CNN** and **MediaPipe hand detection** to enable accurate gesture recognition.
- Built an **end-to-end data processing pipeline** (hand cropping, background removal, grayscale, normalization) and trained the CNN model on **2,515 images** (72/8/20 split) with **5-frame temporal smoothing**, achieving **94% test accuracy** on unseen data.

Autonomous Line-Following Vehicle — *Tampa, Florida*

Design Lead, Team of 5 | C, Arduino, TinkerCAD — *Jan. 2023 - May. 2023*

- **Engineered** an **autonomous** 4-wheel **robotic vehicle** featuring **ultrasonic sensor** integration for **real-time obstacle avoidance**; achieved a **100% collision-free** record across all **20+** test trials.
- **Led** the 5-member team through the full design-to-build lifecycle, implementing **Arduino** control logic in **C** to process sensor data at **10 Hz**, optimizing motor response times and ensuring high-precision line-tracking stability, placing **1st** out of **18 teams**.

Leadership Experience

Vice President — *Tampa, Florida*

Stampede Sounds Collective, University of South Florida — *Apr. 2024 – May. 2025*

- Served as **second-in-command** for a **14-member executive board**, overseeing strategic operations, **20+ annual activities**, and event execution for a **400+** member organization while **deputizing** for the **President** in administrative meetings.